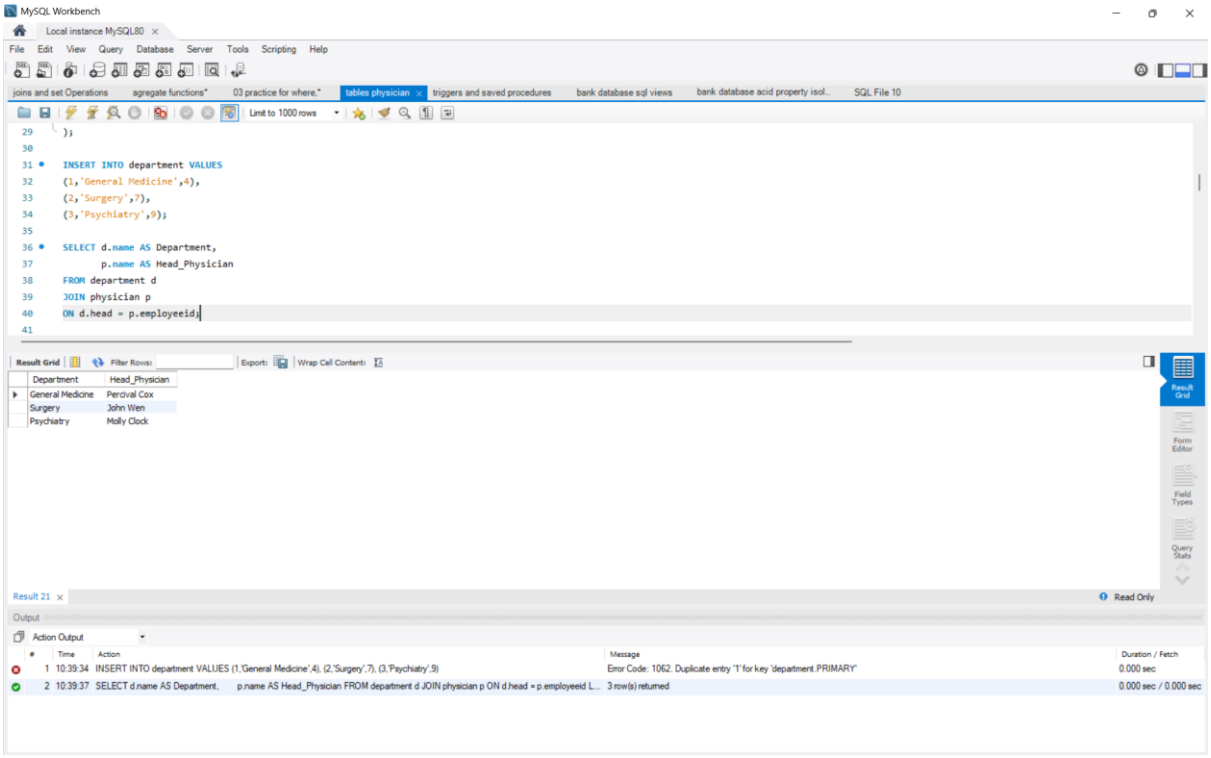
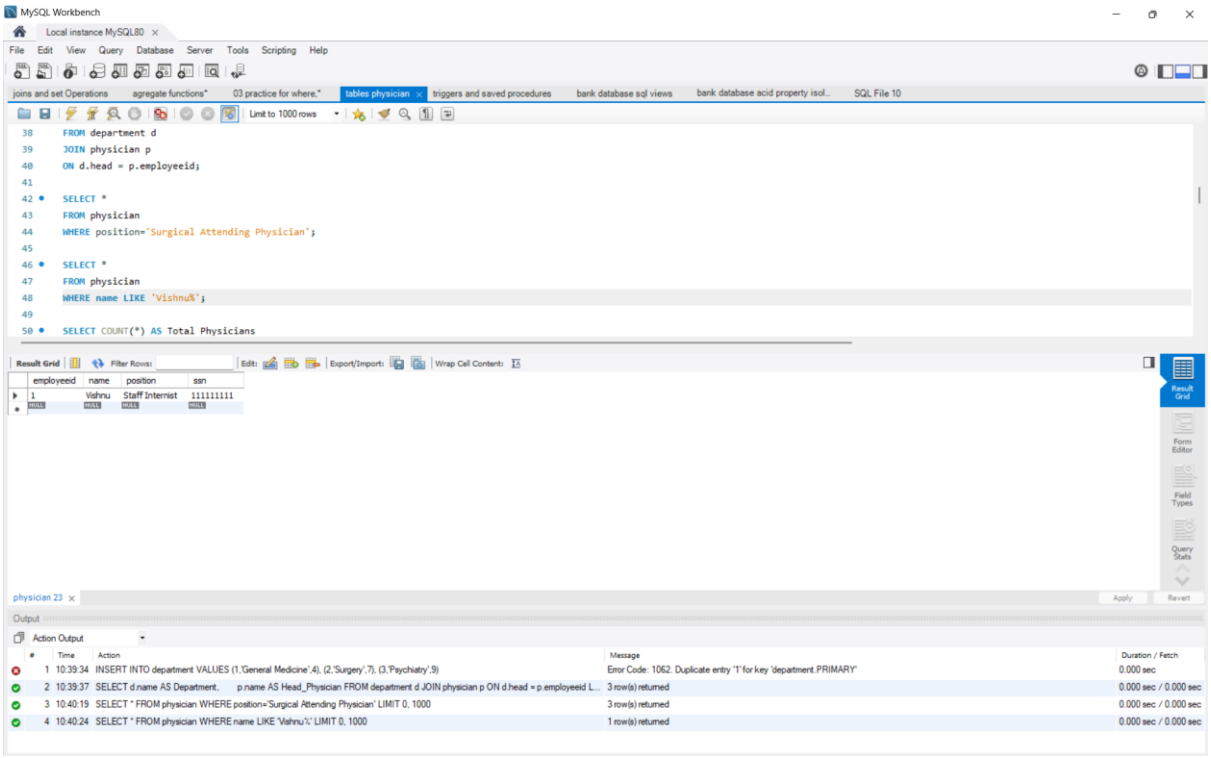
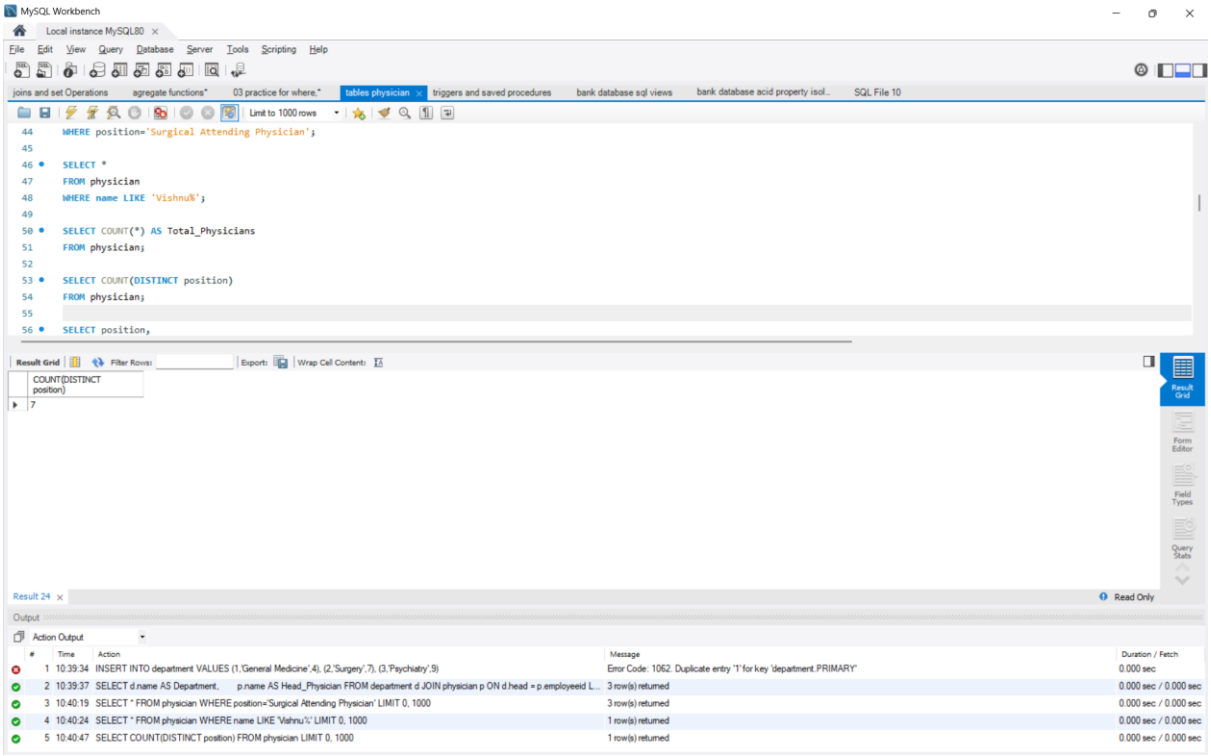
1.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
);
INSERT INTO department VALUES
(1,'General Medicine',4),
(2,'Surgery',7),
(3,'Psychiatry',9);
SELECT d.name AS Department,
p.name AS Head_Physician
FROM department d
JOIN physician p
ON d.head = p.employeeid;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Department
Head_Physician
General Medicine
Percival Cox
Surgery
John Wen
Psychiatry
Molly Clock
Result 21
Read Only
Output
Action Output
Time
Action
Message
Duration / Fetch
1 10:39:34 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)
Error Code: 1062. Duplicate entry '1' for key 'department.PRIMARY'
0.000 sec
2 10:39:37 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid L...
3 row(s) returned
0.000 sec / 0.000 sec

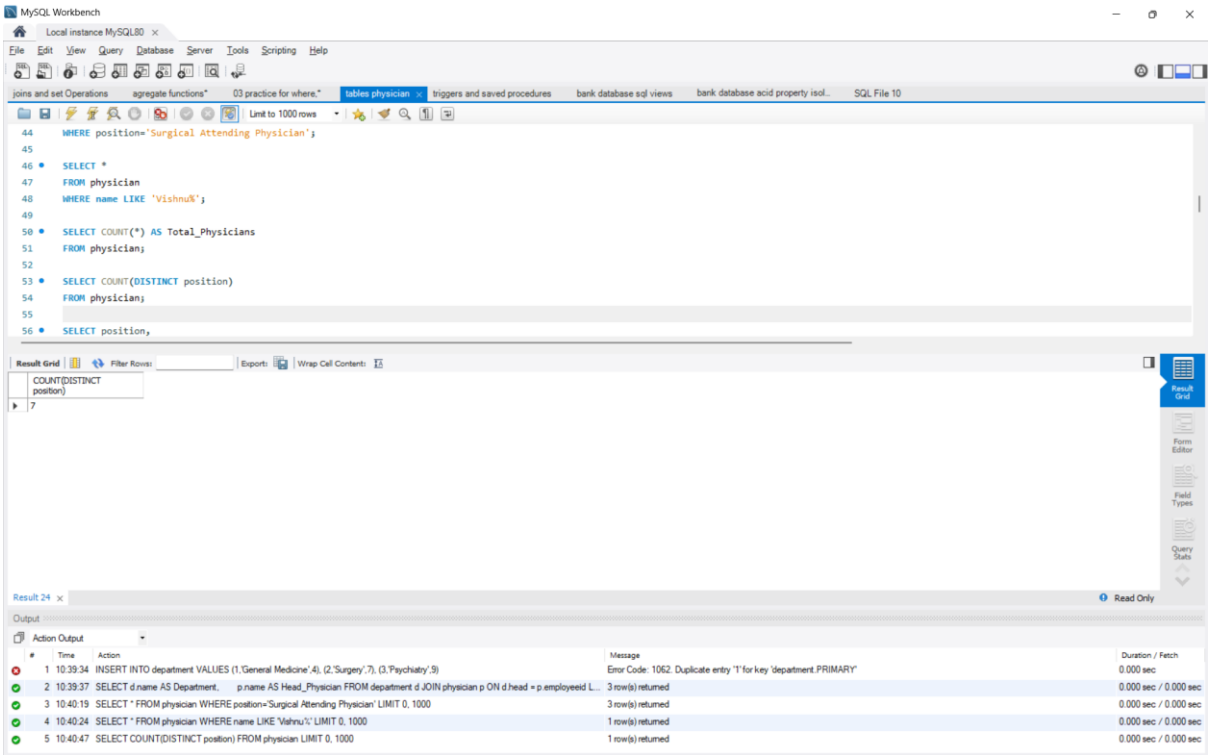
2.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
FROM department d
JOIN physician p
ON d.head = p.employeeid;
SELECT *
FROM physician
WHERE position='Surgical Attending Physician';
SELECT *
FROM physician
WHERE name LIKE 'Vishnu%';
SELECT COUNT(*) AS Total Physicians
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employeeid
name
position
ssn
1
Vishnu
Staff Internist
111111111
physician 23
Apply
Revert
Output
Action Output
Time
Action
Message
Duration / Fetch
1 10:39:34 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9)
Error Code: 1062. Duplicate entry '1' for key 'department.PRIMARY'
0.000 sec
2 10:39:37 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid L...
3 row(s) returned
0.000 sec / 0.000 sec
3 10:40:19 SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000
3 row(s) returned
0.000 sec / 0.000 sec
4 10:40:24 SELECT * FROM physician WHERE name LIKE 'Vishnu%' LIMIT 0, 1000
1 row(s) returned
0.000 sec / 0.000 sec

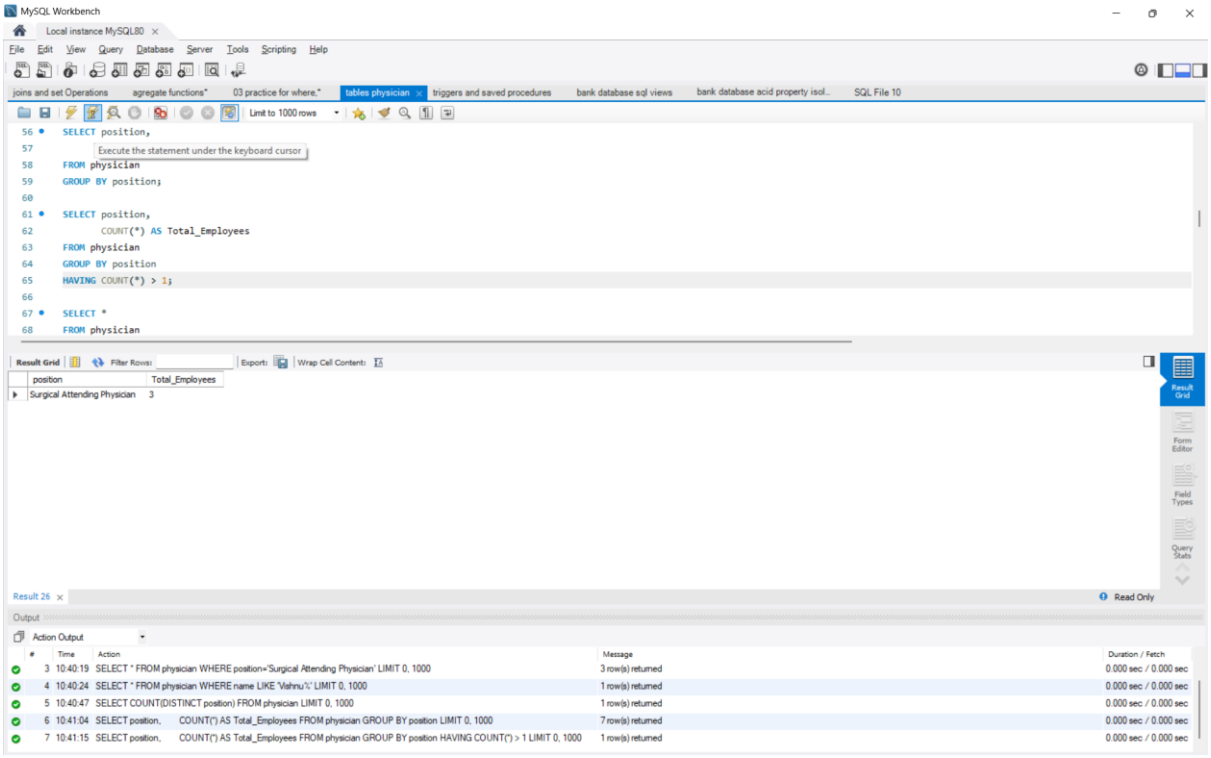
3.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
44 WHERE position='Surgical Attending Physician';
45
46 • SELECT *
47 FROM physician
48 WHERE name LIKE 'Vishnu%';
49
50 • SELECT COUNT(*) AS Total_Physicians
51 FROM physician;
52
53 • SELECT COUNT(DISTINCT position)
54 FROM physician;
55
56 • SELECT position,
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
COUNT(DISTINCT position)
7
Result 24
Read Only
Output
Action Output
Time Action Message Duration / Fetch
1 10:39:34 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) Error Code: 1062. Duplicate entry '1' for key 'department.PRIMARY' 0.000 sec
2 10:39:37 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid L... 3 row(s) returned 0.000 sec / 0.000 sec
3 10:40:19 SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
4 10:40:24 SELECT * FROM physician WHERE name LIKE 'Vishnu%' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 10:40:47 SELECT COUNT(DISTINCT position) FROM physician LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

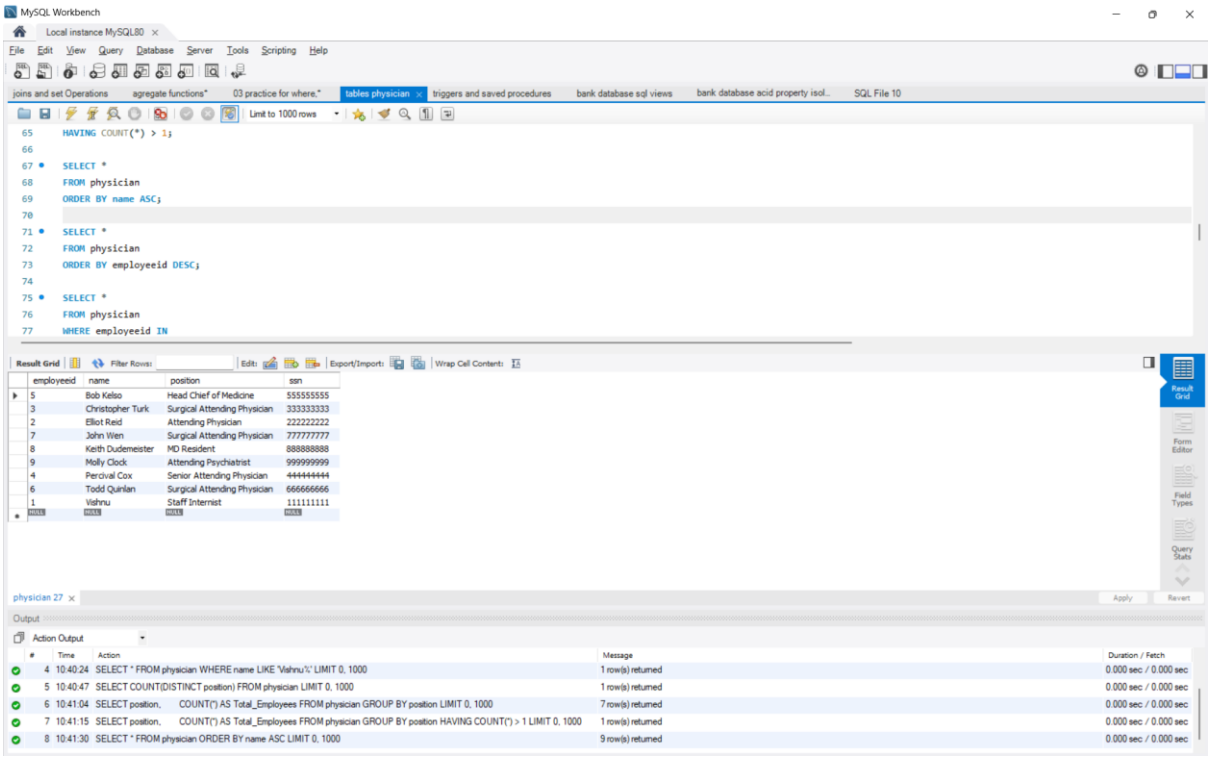
4.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
44 WHERE position='Surgical Attending Physician';
45
46 • SELECT *
47 FROM physician
48 WHERE name LIKE 'Vishnu%';
49
50 • SELECT COUNT(*) AS Total_Physicians
51 FROM physician;
52
53 • SELECT COUNT(DISTINCT position)
54 FROM physician;
55
56 • SELECT position,
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
COUNT(DISTINCT position)
7
Result 24
Read Only
Output
Action Output
Time Action Message Duration / Fetch
1 10:39:34 INSERT INTO department VALUES (1,'General Medicine',4), (2,'Surgery',7), (3,'Psychiatry',9) Error Code: 1062. Duplicate entry '1' for key 'department.PRIMARY' 0.000 sec
2 10:39:37 SELECT d.name AS Department, p.name AS Head_Physician FROM department d JOIN physician p ON d.head = p.employeeid L... 3 row(s) returned 0.000 sec / 0.000 sec
3 10:40:19 SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
4 10:40:24 SELECT * FROM physician WHERE name LIKE 'Vishnu%' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 10:40:47 SELECT COUNT(DISTINCT position) FROM physician LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

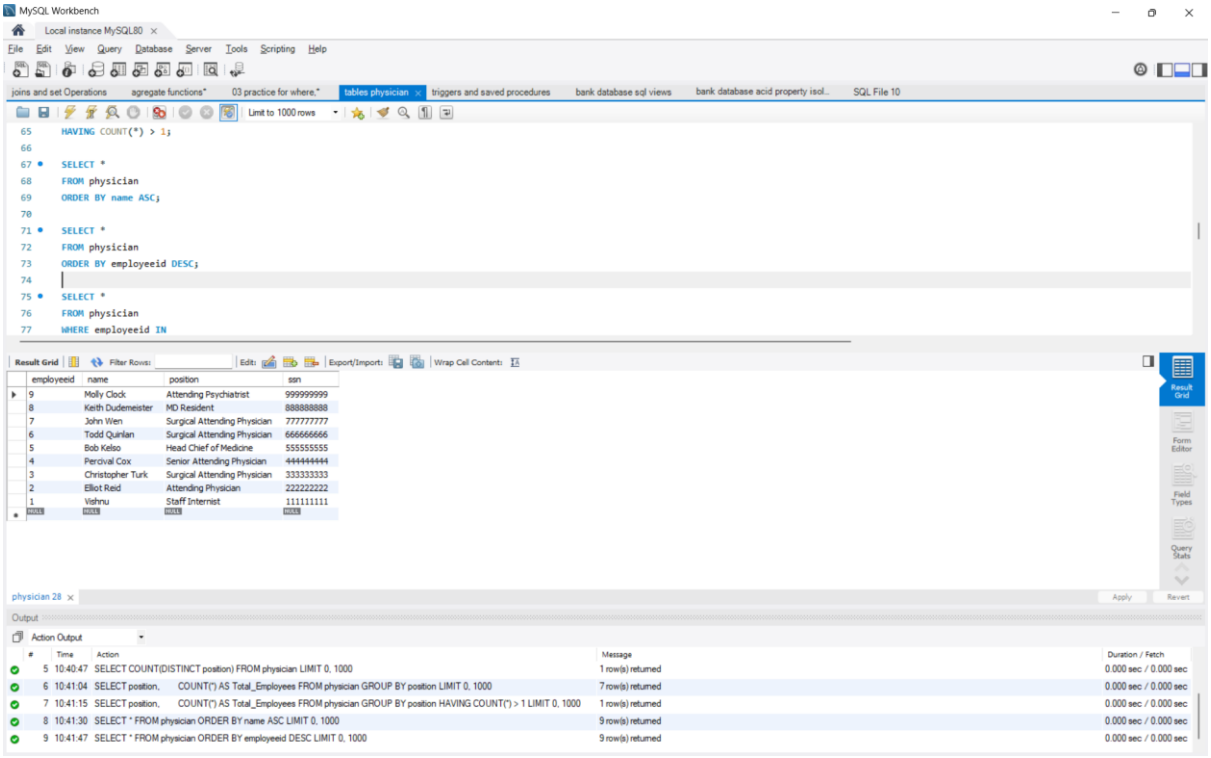
5.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
56 • SELECT position,
57 COUNT(*) AS Total_Employees
58 FROM physician
59 GROUP BY position;
60
61 • SELECT position,
62 COUNT(*) AS Total_Employees
63 FROM physician
64 GROUP BY position
65 HAVING COUNT(*) > 1;
66
67 • SELECT *
68 FROM physician
Execute the statement under the keyboard cursor
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
position
Total_Employees
Surgical Attending Physician 3
Result 26
Read Only
Output
Action Output
Time Action Message Duration / Fetch
3 10:40:19 SELECT * FROM physician WHERE position='Surgical Attending Physician' LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
4 10:40:24 SELECT * FROM physician WHERE name LIKE 'Vishnu%' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 10:40:47 SELECT COUNT(DISTINCT position) FROM physician LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
6 10:41:04 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000 7 row(s) returned 0.000 sec / 0.000 sec
7 10:41:15 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

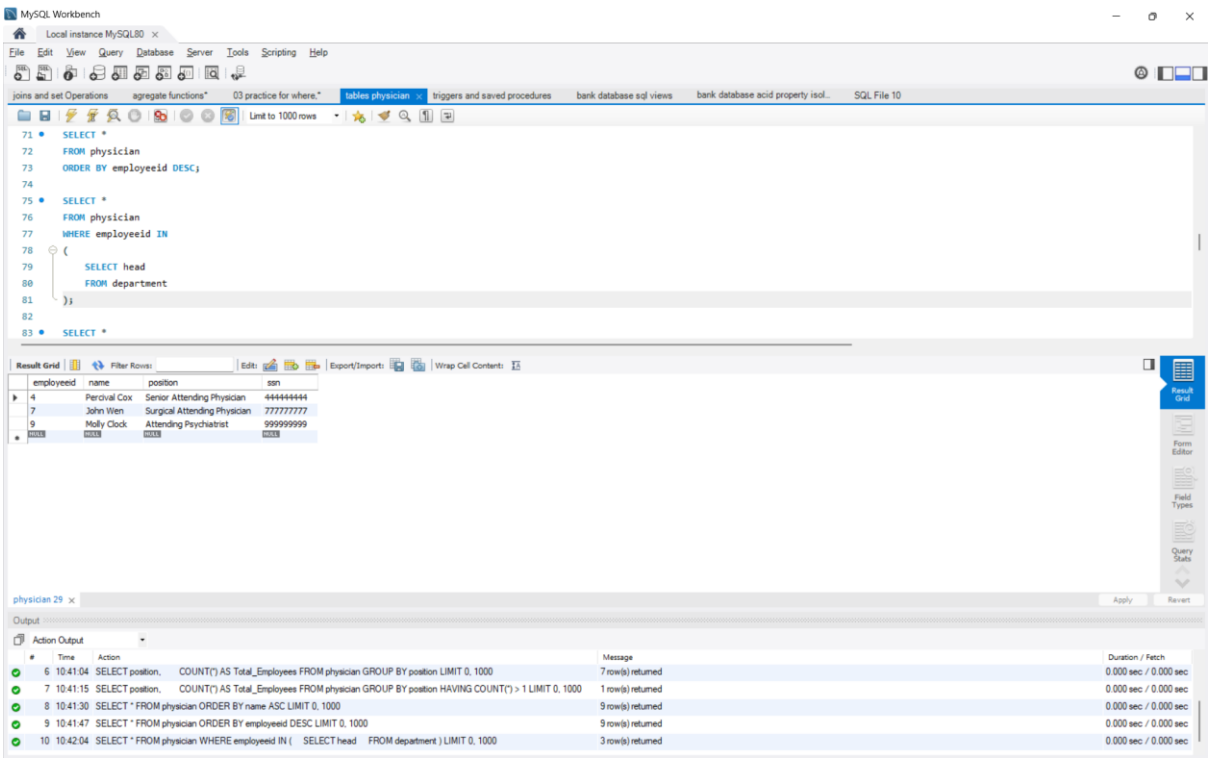
6.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
65 HAVING COUNT(*) > 1;
66
67 • SELECT *
68 FROM physician
69 ORDER BY name ASC;
70
71 • SELECT *
72 FROM physician
73 ORDER BY employeeid DESC;
74
75 • SELECT *
76 FROM physician
77 WHERE employeeid IN
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employeeid name position ssn
5 Bob Kelso Head Chief of Medicine 555555555
3 Christopher Turk Surgical Attending Physician 333333333
2 Elliot Reid Attending Physician 222222222
7 John Wen Surgical Attending Physician 777777777
8 Keith Dudemeister MD Resident 888888888
9 Molly Clock Attending Psychiatrist 999999999
4 Percival Cox Senior Attending Physician 444444444
6 Todd Quinlan Surgical Attending Physician 666666666
1 Vishnu Staff Internist 111111111
physician 27
Apply
Revert
Output
Action Output
Time Action Message Duration / Fetch
4 10:40:24 SELECT * FROM physician WHERE name LIKE 'Vishnu%' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
5 10:40:47 SELECT COUNT(DISTINCT position) FROM physician LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
6 10:41:04 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000 7 row(s) returned 0.000 sec / 0.000 sec
7 10:41:15 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
8 10:41:30 SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec

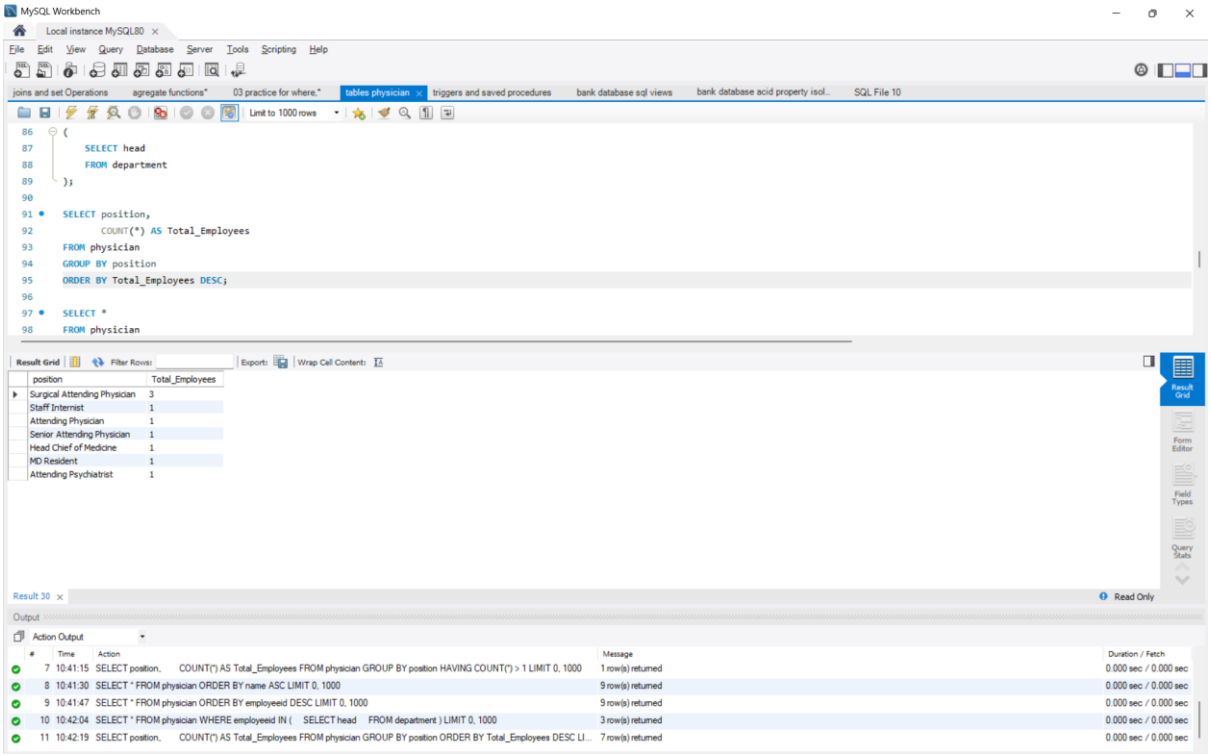
7.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
65 HAVING COUNT(*) > 1;
66
67 • SELECT *
68 FROM physician
69 ORDER BY name ASC;
70
71 • SELECT *
72 FROM physician
73 ORDER BY employeeid DESC;
74
75 • SELECT *
76 FROM physician
77 WHERE employeeid IN
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employeeid name position ssn
9 Molly Clock Attending Psychiatrist 999999999
8 Keith Dudemeister MD Resident 888888888
7 John Wen Surgical Attending Physician 777777777
6 Todd Quinlan Surgical Attending Physician 666666666
5 Bob Kelso Head Chief of Medicine 555555555
4 Percival Cox Senior Attending Physician 444444444
3 Christopher Turk Surgical Attending Physician 333333333
2 Elliot Reid Attending Physician 222222222
1 Vishnu Staff Internist 111111111
physician 28
Output
Action Output
Time Action Message Duration / Fetch
5 10:40:47 SELECT COUNT(DISTINCT position) FROM physician LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
6 10:41:04 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000 7 row(s) returned 0.000 sec / 0.000 sec
7 10:41:15 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
8 10:41:30 SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec
9 10:41:47 SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec

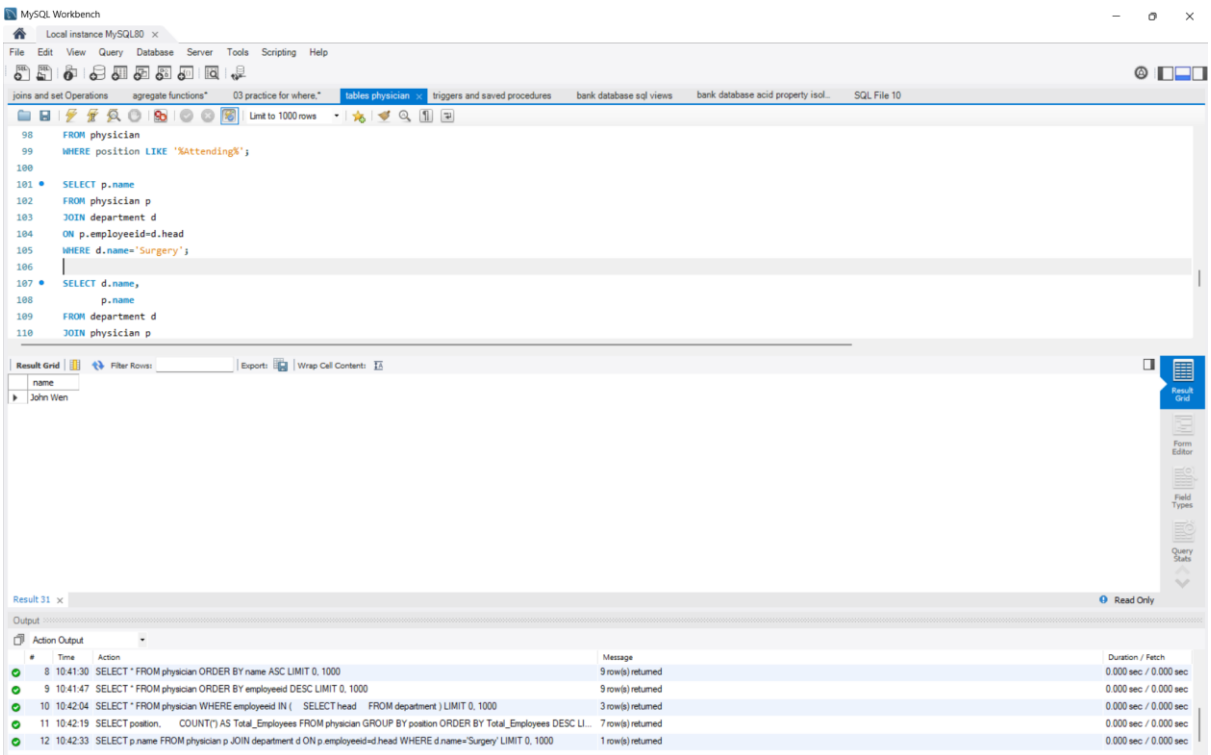
8.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
71 • SELECT *
72 FROM physician
73 ORDER BY employeeid DESC;
74
75 • SELECT *
76 FROM physician
77 WHERE employeeid IN
78 (
79 SELECT head
80 FROM department
81);
82
83 • SELECT *
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employeeid name position ssn
4 Percival Cox Senior Attending Physician 444444444
7 John Wen Surgical Attending Physician 777777777
9 Molly Clock Attending Psychiatrist 999999999
physician 29
Output
Action Output
Time Action Message Duration / Fetch
6 10:41:04 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position LIMIT 0, 1000 7 row(s) returned 0.000 sec / 0.000 sec
7 10:41:15 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
8 10:41:30 SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec
9 10:41:47 SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec
10 10:42:04 SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec

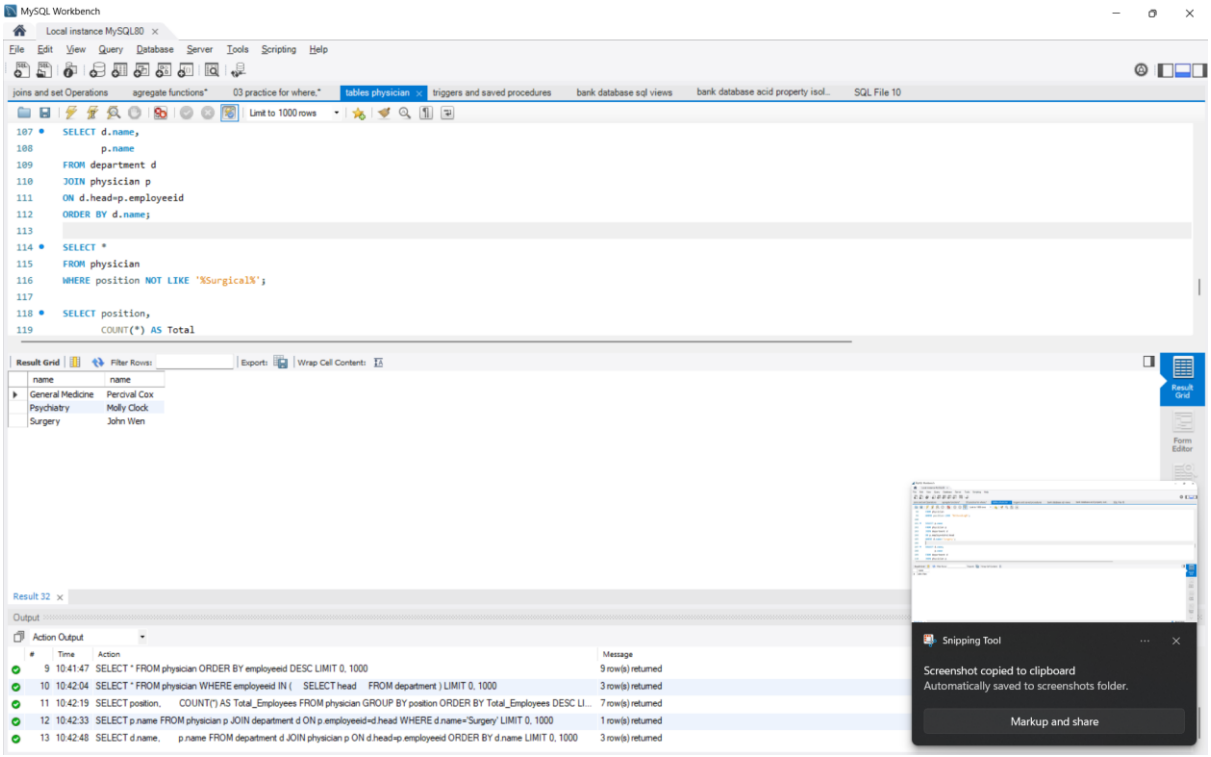
9.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
86 (
87 SELECT head
88 FROM department
89);
90
91 SELECT position,
92 COUNT(*) AS Total_Employees
93 FROM physician
94 GROUP BY position
95 ORDER BY Total_Employees DESC;
96
97 SELECT *
98 FROM physician
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
position Total_Employees
Surgical Attending Physician 3
Staff Internist 1
Attending Physician 1
Senior Attending Physician 1
Head Chief of Medicine 1
MD Resident 1
Attending Psychiatrist 1
Result 30
Read Only
Output
Action Output
Time Action Message Duration / Fetch
7 10:41:15 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position HAVING COUNT(*) > 1 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
8 10:41:30 SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec
9 10:41:47 SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec
10 10:42:04 SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
11 10:42:19 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total_Employees DESC LI... 7 row(s) returned 0.000 sec / 0.000 sec

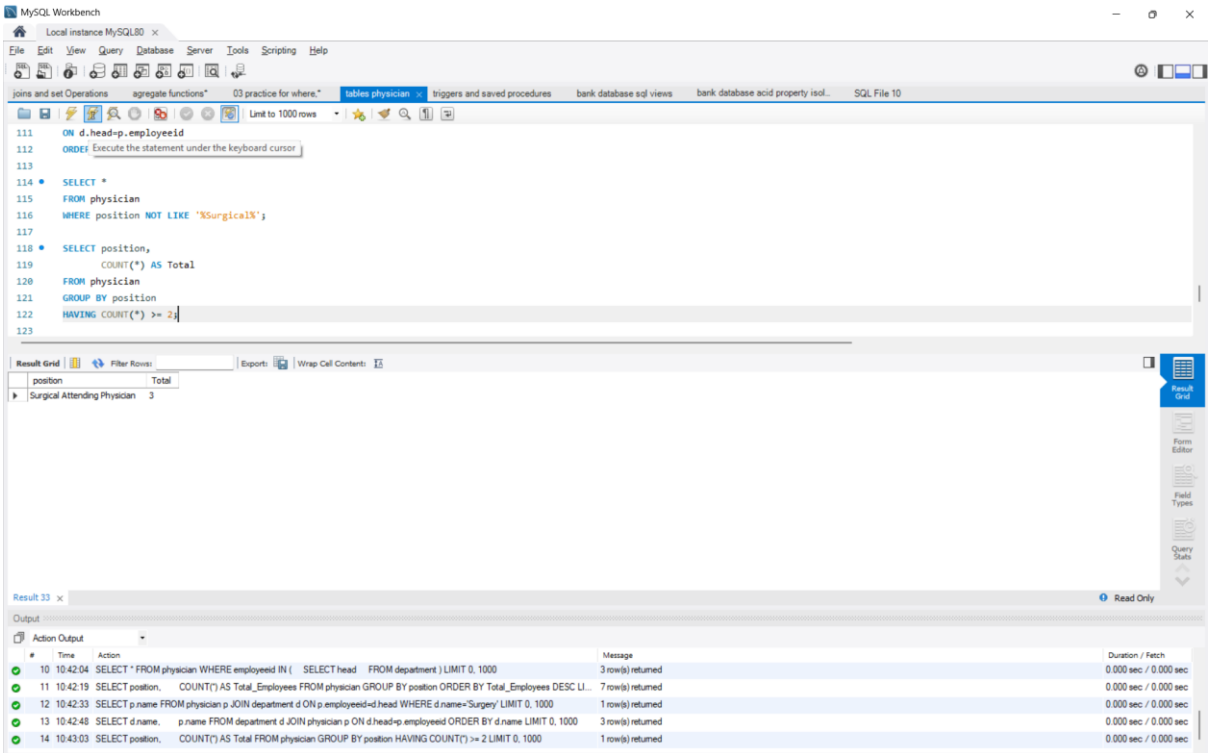
10.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
98 FROM physician
99 WHERE position LIKE '%Attending%';
100
101 SELECT p.name
102 FROM physician p
103 JOIN department d
104 ON p.employeeid=d.head
105 WHERE d.name='Surgery';
106
107 SELECT d.name,
108 p.name
109 FROM department d
110 JOIN physician p
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
name
John Wen
Result 31
Read Only
Output
Action Output
Time Action Message Duration / Fetch
8 10:41:30 SELECT * FROM physician ORDER BY name ASC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec
9 10:41:47 SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000 9 row(s) returned 0.000 sec / 0.000 sec
10 10:42:04 SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
11 10:42:19 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total_Employees DESC LI... 7 row(s) returned 0.000 sec / 0.000 sec
12 10:42:33 SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

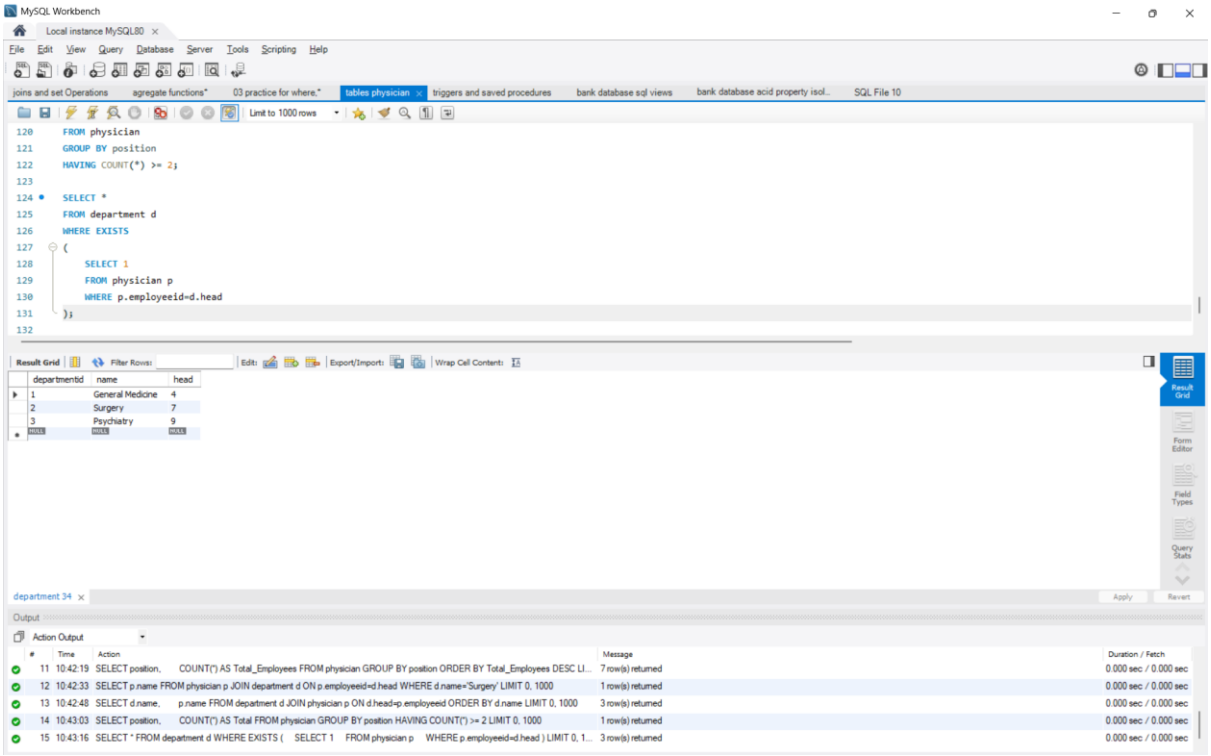
11.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol..
SQL File 10
Limit to 1000 rows
SELECT d.name,
p.name
FROM department d
JOIN physician p
ON d.head=p.employeeid
ORDER BY d.name;
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
SELECT position,
COUNT(*) AS Total
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
name name
General Medicine Percival Cox
Psychiatry Molly Clock
Surgery John Wen
Result 32
Output
Action Output
Time Action Message
9 10:41:47 SELECT * FROM physician ORDER BY employeeid DESC LIMIT 0, 1000 9 row(s) returned
10 10:42:04 SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000 3 row(s) returned
11 10:42:19 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total_Employees DESC LI... 7 row(s) returned
12 10:42:33 SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery' LIMIT 0, 1000 1 row(s) returned
13 10:42:48 SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d.name LIMIT 0, 1000 3 row(s) returned
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Markup and share

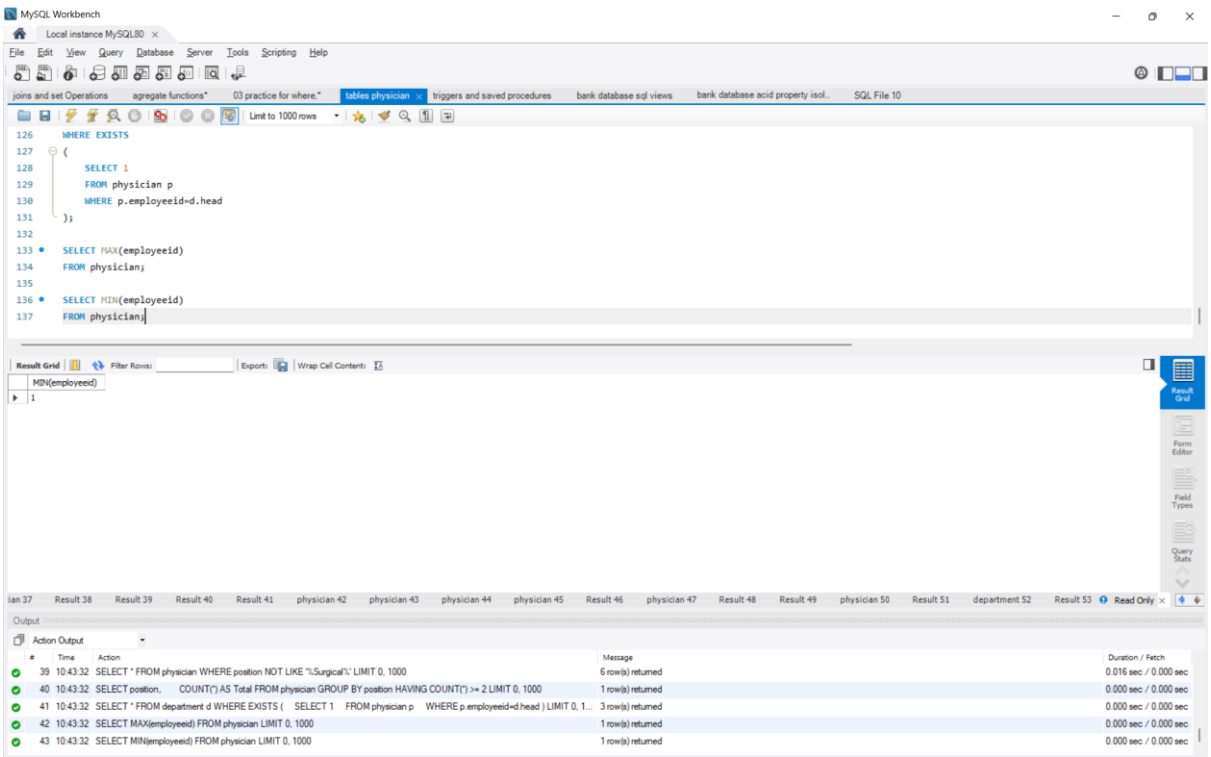
12.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol..
SQL File 10
Limit to 1000 rows
ON d.head=p.employeeid
ORDER BY d.name;
Execute the statement under the keyboard cursor
SELECT *
FROM physician
WHERE position NOT LIKE '%Surgical%';
SELECT position,
COUNT(*) AS Total
FROM physician
GROUP BY position
HAVING COUNT(*) >= 2;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
position Total
Surgical Attending Physician 3
Result 33
Read Only
Output
Action Output
Time Action Message Duration / Fetch
10 10:42:04 SELECT * FROM physician WHERE employeeid IN (SELECT head FROM department) LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
11 10:42:19 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total_Employees DESC LI... 7 row(s) returned 0.000 sec / 0.000 sec
12 10:42:33 SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
13 10:42:48 SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d.name LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
14 10:43:03 SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec

13.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
120 FROM physician
121 GROUP BY position
122 HAVING COUNT(*) >= 2;
123
124 SELECT *
125 FROM department d
126 WHERE EXISTS
127 (
128 SELECT 1
129 FROM physician p
130 WHERE p.employeeid=d.head
131);
132
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
departmentid name head
1 General Medicine 4
2 Surgery 7
3 Psychiatry 9
department 34
Apply
Revert
Output
Action Output
Time Action Message Duration / Fetch
11 10:42:19 SELECT position, COUNT(*) AS Total_Employees FROM physician GROUP BY position ORDER BY Total_Employees DESC LI... 7 row(s) returned 0.000 sec / 0.000 sec
12 10:42:33 SELECT p.name FROM physician p JOIN department d ON p.employeeid=d.head WHERE d.name='Surgery' LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
13 10:42:48 SELECT d.name, p.name FROM department d JOIN physician p ON d.head=p.employeeid ORDER BY d.name LIMIT 0, 1000 3 row(s) returned 0.000 sec / 0.000 sec
14 10:43:03 SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
15 10:43:16 SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid=d.head) LIMIT 0, 1... 3 row(s) returned 0.000 sec / 0.000 sec

14.


MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
joins and set Operations
agregate functions*
03 practice for where.*
tables physician
triggers and saved procedures
bank database sql views
bank database acid property isol...
SQL File 10
Limit to 1000 rows
126 WHERE EXISTS
127 (
128 SELECT 1
129 FROM physician p
130 WHERE p.employeeid=d.head
131);
132
133 SELECT MAX(employeeid)
134 FROM physician;
135
136 SELECT MIN(employeeid)
137 FROM physician;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
MIN(employeeid)
1
Result 38 Result 39 Result 40 Result 41 physician 42 physician 43 physician 44 physician 45 Result 46 physician 47 Result 48 Result 49 physician 50 Result 51 department 52 Result 53
Read Only
Output
Action Output
Time Action Message Duration / Fetch
39 10:43:32 SELECT * FROM physician WHERE position NOT LIKE '%Surgical%' LIMIT 0, 1000 6 row(s) returned 0.016 sec / 0.000 sec
40 10:43:32 SELECT position, COUNT(*) AS Total FROM physician GROUP BY position HAVING COUNT(*) >= 2 LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
41 10:43:32 SELECT * FROM department d WHERE EXISTS (SELECT 1 FROM physician p WHERE p.employeeid=d.head) LIMIT 0, 1... 3 row(s) returned 0.000 sec / 0.000 sec
42 10:43:32 SELECT MAX(employeeid) FROM physician LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec
43 10:43:32 SELECT MIN(employeeid) FROM physician LIMIT 0, 1000 1 row(s) returned 0.000 sec / 0.000 sec